Model Speed Benchmark Results

📊 Executive Summary

MODEL	SILO TPS	LISO TPS	BALANCED TPS	KEY OBSERVATION
nemotron-mini-4b-instruct	101.6	56.4	76.8	Solid all-rounder; LISO throughput is inconsistent
llama-3.3-nemotron-super-49b-v1	16.3	24.4	21.5	Slowest throughput; startup time highly variable
nemotron-3-nano-omni-30b-a3b-reasoning	**317.5**	**872.2**	**335.8**	🏆 Fastest across all scenarios by a wide margin
qwen3-next-80b-a3b-instruct	30.6	40.1	31.0	Mid-range speed; startup time unreliable
qwen3.5-122b-a10b	16.8	31.6	26.9	Most inconsistent; treat all figures as preliminary

> ***TPS = tokens per second during generation (higher is better). All figures are mean values.***

> ***Methodology:*** *Single-request / load-isolated. TPS =* `output_tokens ÷ (E2E − TTFT)` *(decode phase only).* ***Samples:*** *3 valid runs per cell (anomalous runs auto-discarded and retried).* ***Token counter:*** *tiktoken / cl100k_base (falls back to API-reported count).* ***Scenarios:*** *SILO = Short In Long Out · LISO = Long In Short Out · BALANCED = medium prompt/output*

Results Table

MODEL	SCENARIO	TTFT MEAN ± STD (S)	TPS MEAN ± STD	E2E TIME (S)	TOKENS	STABILITY NOTES
nemotron-mini-4b-instruct	SILO	0.730 ± 0.236	101.6 ± 1.1	7.22	659	✅ Stable
nemotron-mini-4b-instruct	LISO	0.764 ± 0.153	56.4 ± 45.6	5.93	120	⚠ High TPS variance (CV 81%)
nemotron-mini-4b-instruct	BALANCED	0.684 ± 0.079	76.8 ± 24.8	6.10	277	✅ Stable
llama-3.3-nemotron-super-49b-v1	SILO	0.957 ± 0.180	16.3 ± 3.0	56.10	884	✅ Stable
llama-3.3-nemotron-super-49b-v1	LISO	3.939 ± 4.881	24.4 ± 0.9	8.86	120	⚠ High TTFT variance
llama-3.3-nemotron-super-49b-v1	BALANCED	8.893 ± 13.592	21.5 ± 4.9	22.34	280	⚠ High TTFT variance
nemotron-3-nano-omni-30b-a3b-reasoning	SILO	1.285 ± 0.272	317.5 ± 72.3	4.21	900	⚠ High TPS variance (CV 23%)
nemotron-3-nano-omni-30b-a3b-reasoning	LISO	1.024 ± 0.032	872.2 ± 78.2	1.16	120	✅ Stable
nemotron-3-nano-omni-30b-a3b-reasoning	BALANCED	0.993 ± 0.086	335.8 ± 69.8	1.85	280	⚠ High TPS variance (CV 21%)
qwen3-next-80b-a3b-instruct	SILO	3.845 ± 4.951	30.6 ± 1.6	31.63	850	⚠ High TTFT variance
qwen3-next-80b-a3b-instruct	LISO	3.380 ± 1.168	40.1 ± 11.8	6.54	120	✅ Stable

MODEL	SCENARIO	TTFT MEAN ± STD (S)	TPS MEAN ± STD	E2E TIME (S)	TOKENS	STABILITY NOTES
qwen3-next-80b-a3b-instruct	BALANCED	6.062 ± 4.322	31.0 ± 10.7	15.27	257	⚠ High TTFT variance
qwen3.5-122b-a10b	SILO	13.541 ± 9.215	16.8 ± 10.4	81.16	878	⚠ High TTFT & TPS variance
qwen3.5-122b-a10b	LISO	7.649 ± 5.956	31.6 ± 53.0	94.24	120	⚠ High TTFT & TPS variance
qwen3.5-122b-a10b	BALANCED	13.659 ± 1.236	26.9 ± 26.8	47.23	264	⚠ High TPS variance (CV 100%)

🏆 Fastest Model by Scenario

SCENARIO	FASTEST MODEL	TPS MEAN ± STD	TTFT MEAN (S)
SILO	`nemotron-3-nano-omni-30b-a3b-reasoning`	317.5 ± 72.3 tok/s	1.285 s
LISO	`nemotron-3-nano-omni-30b-a3b-reasoning`	872.2 ± 78.2 tok/s	1.024 s
BALANCED	`nemotron-3-nano-omni-30b-a3b-reasoning`	335.8 ± 69.8 tok/s	0.993 s

> ***Omni note:*** *The Omni model leads across all three scenarios. TPS variance is elevated on SILO (CV 23%) and BALANCED (CV 21%), so treat those figures as approximate pending additional samples.*

⚠ Instability & Anomaly Notes

nemotron-mini-4b-instruct — LISO: High TPS Variance

TPS std of 45.6 against a mean of 56.4 gives a CV of ~81%, well above the 40% instability threshold.

- **Recommendation:** Increase `N_SAMPLES` to 5–7 for this model/scenario to obtain a reliable p95 estimate.

llama-3.3-nemotron-super-49b-v1 — LISO & BALANCED: High TTFT Variance

TTFT std exceeds the mean in both scenarios (LISO: 4.881 s std vs. 3.939 s mean; BALANCED: 13.592 s std vs. 8.893 s mean), indicating significant server-side queueing or cold-start variability.

- **Recommendation:** Re-run during off-peak hours and increase `N_SAMPLES`; treat current TTFT figures as approximate.

nemotron-3-nano-omni-30b-a3b-reasoning — High TPS Variance (SILO / BALANCED)

LISO and BALANCED TPS=0 anomalies from earlier runs are resolved; valid TPS is now recorded across all three scenarios. SILO (CV 23%) and BALANCED (CV 21%) remain below the 40% instability threshold, though the run-to-run variability in decode throughput is worth monitoring.

- **Recommendation:** Increase `N_SAMPLES` to 5–7 for SILO and BALANCED to tighten confidence intervals.

qwen3-next-80b-a3b-instruct — High TTFT Variance (SILO / BALANCED)

TTFT std exceeds the mean on SILO (4.951 s std vs. 3.845 s mean), indicating significant server-side queueing or cold-start variability. BALANCED is similarly unstable (4.322 s std vs. 6.062 s mean).

- **Recommendation:** Re-run during off-peak hours and increase `N_SAMPLES`; treat current TTFT figures as approximate.

qwen3.5-122b-a10b — High TTFT & TPS Variance (All Scenarios)

All three scenarios show extreme variability. TTFT std exceeds the mean on SILO (9.215 s vs. 13.541 s) and LISO (5.956 s vs. 7.649 s). TPS CV reaches ~62% on SILO, ~168% on LISO, and ~100% on BALANCED — all well above the 40% instability threshold. E2E times are also substantially higher than prior runs (SILO 81.16 s, LISO 94.24 s), suggesting severe server-side load or queueing pressure at time of measurement.

- **Recommendation:** Treat all qwen3.5-122b-a10b figures as preliminary. Re-run during off-peak hours with `N_SAMPLES` increased to 7–10.

NVIDIA UI vs. API Comparison — Omni Model (SILO)

SOURCE	TPS OBSERVED	NOTES
NVIDIA UI — Run 1	129.98 tok/s	
NVIDIA UI — Run 2	142.72 tok/s	
NVIDIA UI — Run 3	213.97 tok/s	
NVIDIA UI — Run 4	299.27 tok/s	
NVIDIA UI Range	**129.98 – 299.27 tok/s**	High run-to-run variance
API Benchmark (SILO)	**317.5 ± 72.3 tok/s**	Decode-only TPS

Key observations:

- API TPS (317.5) slightly exceeds the upper bound of the UI-observed range (~299). The difference is expected: the API benchmark measures only the token generation step, while the UI figure includes rendering overhead, session warm-up effects, and variable request queuing.
- UI run-to-run variance (129.98 → 299.27 tok/s, a 2.3× spread across 4 runs) underscores the need for repeated sampling and anomaly filtering.
- The API SILO TPS std of 72.3 (CV 23%) is similarly elevated, suggesting that server-side load variability affects both measurement paths.

Benchmark Configuration

PARAMETER	VALUE
Samples (target valid runs)	3 per (model × scenario)
Max attempts per cell	6
Temperature	0.2
Min output tokens (anomaly floor)	5
Min TPS (anomaly floor)	0.5 tok/s
Instability CV threshold	40%
Token counter	tiktoken / cl100k_base
Connect timeout	15 s
Read timeout	180 s